

Designation: D 4892-24

Standard Specification for Ultra-High Molecular Weight Polyethylene (UHMWPE) Sheet

This standard was issued on February 10, 2024; the version of record is continuously maintained.

1. Scope

1.1 This specification covers ultra-high molecular weight polyethylene sheets for wear-resistant and impact-resistant applications

1.2 The UHMWPE sheets covered by this specification are intended for use in manufacturing of conveyor components, chute liners, bearing surfaces, and machine parts requiring low friction and high abrasion resistance.

1.3 The values stated in SI units are to be regarded as standard. No other units of measurement are included in this standard.

1.4 This standard does not purport to address all of the safety concerns, if any, associated with its use. It is the responsibility of the user of this standard to establish appropriate safety, health, and environmental practices and determine the applicability of regulatory limitations prior to use.

2. Referenced Documents

2.1 ASTM Standards:

D 792 Test Methods for Density and Specific Gravity of Plastics

D 638 Test Method for Tensile Properties of Plastics

D 4020 Test Method for Ultra-High-Molecular-Weight Polyethylene Molding and Extrusion Materials

D 1894 Test Method for Static and Kinetic Coefficients of Friction of Plastic Film

3. Terminology

3.1 Definitions:

3.1.1 ultra-high molecular weight—polyethylene with molecular weight greater than 3 million g/mol

3.1.2 virgin material—polymer that has not been previously processed or recycled

3.1.3 coefficient of friction—the ratio of the force of friction between two bodies

4. Classification

4.1 UHMWPE sheets shall be classified according to thickness and molecular weight grade.

4.1.1 **Type A**—Molecular weight 3-6 million g/mol, thickness 6-25 mm

4.1.2 **Type B**—Molecular weight 6-9 million g/mol, thickness 25-100 mm

4.1.3 **Type C**—Molecular weight > 9 million g/mol, custom thickness

5. Ordering Information

5.1 Orders for material under this specification shall include the following information:

5.1.1 ASTM designation and year of issue

5.1.2 Type (A, B, or C)

5.1.3 Sheet dimensions (length × width × thickness in mm)

5.1.4 Quantity (number of sheets or total area in m²)

5.1.5 Color (natural white, black, or custom)

6. Materials and Manufacture

6.1 Sheets shall be manufactured from virgin UHMWPE resin meeting molecular weight requirements.

6.2 No reground or recycled material shall be used without prior approval.

6.3 Colorants and additives, if used, shall not adversely affect mechanical properties.

7. Mechanical and Physical Requirements

Property	Requirement	Test Method
Density	0.930-0.945 g/cm ³	ASTM D 792
Tensile strength	≥ 21 MPa	ASTM D 638
Elongation at break	≥ 300%	ASTM D 638
Coefficient of friction	≤ 0.20	ASTM D 1894

8. Dimensions and Permissible Variations

8.1 Standard sheet sizes are 1220 mm × 2440 mm. Custom sizes available upon request.

8.2 Thickness tolerance shall be ±10% for sheets up to 50 mm thick.

8.3 For sheets greater than 50 mm thick, tolerance shall be ±15%.

8.4 Flatness deviation shall not exceed 5 mm per meter.

9. Workmanship, Finish, and Appearance

9.1 Sheets shall have uniform color and be free from cracks, voids, foreign inclusions, and surface defects that would impair function.

10. Sampling

10.1 One test specimen per production lot shall be sampled for property verification. Lot size shall not exceed 5000 kg.

11. Test Methods

11.1 **Density Measurement**—Determine density using water displacement method per ASTM D 792.

11.2 **Tensile Testing**—Machine specimens per ASTM D 638 and test at standard conditions.

11.3 **Friction Testing**—Measure static and kinetic friction coefficients per ASTM D 1894 against polished steel.

12. Inspection

12.1 Inspection of the material shall be made as agreed upon between the purchaser and the seller as part of the purchase contract.

13. Rejection and Rehearing

13.1 Material that fails to conform to the requirements of this specification may be rejected. Rejection should be reported to the producer or supplier promptly and in writing.

14. Certification

14.1 A manufacturer's or supplier's certification shall be furnished to the purchaser that the material was manufactured, sampled, tested, and inspected in accordance with this specification and has met the requirements.

15. Product Marking

15.1 Each sheet shall be labeled with manufacturer name, material type, production date, and lot number on protective film.

16. Packaging and Package Marking

16.1 Sheets shall be wrapped in protective polyethylene film and stacked on pallets with corner protection. Moisture barriers are not required.

17. Keywords

17.1 UHMWPE; polyethylene; wear resistant; low friction; impact resistant; sheet